

Module: Msf::Util::DotNetDeserialization — Documentation by YARD 0.9.37

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Content

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Module: Msf::Util::DotNetDeserialization — Documentation by YARD 0.9.37

Module: Msf::Util::DotNetDeserialization

Defined in: lib/msf/util/dot_net_deserialization.rb,
lib/msf/util/dot_net_deserialization/enums.rb, lib/msf/util/dot_net_deserialization/types.rb,
lib/msf/util/dot_net_deserialization/assemblies.rb,
lib/msf/util/dot_net_deserialization/formatters.rb,
lib/msf/util/dot_net_deserialization/gadget_chains.rb,
lib/msf/util/dot_net_deserialization/types/general.rb,
lib/msf/util/dot_net_deserialization/types/primitives.rb,
lib/msf/util/dot_net_deserialization/types/record_values.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/data_set.rb,
lib/msf/util/dot_net_deserialization/types/common_structures.rb,
lib/msf/util/dot_net_deserialization/formatters/los_formatter.rb,
lib/msf/util/dot_net_deserialization/formatters/soap_formatter.rb,
lib/msf/util/dot_net_deserialization/formatters/binary_formatter.rb,
lib/msf/util/dot_net_deserialization/formatters/json_net_formatter.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/claims_principal.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/windows_identity.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/data_set_type_spoof.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/object_data_provider.rb,

lib/msf/util/dot_net_deserialization/gadget_chains/type_confuse_delegate.rb,
lib/msf/util/dot_net_deserialization/gadget_chains/text_formatting_run_properties.rb

Overview

Much of this code is based on the YSoSerial.Net project see: github.com/pwntester/ysoserial.net

Defined Under Namespace

Modules: [Assemblies](#), [Enums](#), [Formatters](#), [GadgetChains](#), [Types](#)

Constant Summary [collapse]()

DEFAULT_FORMATTER =

```
:BinaryFormatter
```

DEFAULT_GADGET_CHAIN =

```
:TextFormattingRunProperties
```

Class Method Summary [collapse]()

- **[.encode_7bit_int(int) ⇒ Object]()**
- **[.formatter_compatible_gadget_chains(formatter) ⇒ Array<Symbol>]()**

Get a list of gadget chains that are compatible with the specified formatter.

- **[.generate(cmd, gadget_chain: DEFAULT_GADGET_CHAIN, formatter: DEFAULT_FORMATTER) ⇒ String]()**

Generates a .NET deserialization payload for the specified OS command using a selected gadget-chain and formatter combination.

- **[.generate_formatted(stream, formatter: DEFAULT_FORMATTER) ⇒ String]()**

Take the specified serialized blob and encapsulate it with the specified formatter.

- **[.generate_gadget_chain(cmd, gadget_chain: DEFAULT_GADGET_CHAIN) ⇒ Types::SerializedStream]()**

Generate a serialized data blob using the specified gadget chain to execute the OS command.

- **[.get_ancestor(obj, ancestor_type, required: true) ⇒ Object]()**

Class Method Details

.encode_7bit_int(int) ⇒ Object

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```

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```
# File 'lib/msf/util/dot_net_deserialization.rb', line 14

def self.encode_7bit_int(int)
  return "\x00".b if int == 0

  encoded_int = []
  while int > 0
    value = int & 0x7f
    int >>= 7
    value |= 0x80 if int > 0
    encoded_int << value
  end

  encoded_int.pack('C*')
end
```

||

.formatter_compatible_gadget_chains(formatter) ⇒ Array<Symbol>

Get a list of gadget chains that are compatible with the specified formatter.

Parameters:

- `formatter (Symbol)` —

The formatter to get gadget chains for.

Returns:

- `(Array<Symbol>)`

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```

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```
# File 'lib/msf/util/dot_net_deserialization.rb', line 87

def self.formatter_compatible_gadget_chains(formatter)
  case formatter
  when :BinaryFormatter, :LosFormatter
    chains = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [NAMES]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html#NAMES-
    constant).select { |name| [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html).const_
    get(name) <= ([Types]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Types.html)::
    [SerializedStream]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Types/SerializedStream.ht
    ml)) }
  when :JsonNetFormatter
    chains = %i[ ObjectDataProvider ]
  when :SoapFormatter
    chains = %i[ ClaimsPrincipal TextFormattingRunProperties WindowsIdentity ]
  else
    raise NotImplementedError, 'The specified formatter is not implemented'
  end

  chains
end
```

||

.generate(cmd, gadget_chain: DEFAULT_GADGET_CHAIN, formatter: DEFAULT_FORMATTER) ⇒ String

Generates a .NET deserialization payload for the specified OS command using a selected gadget-chain and formatter combination.

Parameters:

- cmd (String) —

The OS command to execute.

- gadget_chain (Symbol) (defaults to: *DEFAULT_GADGET_CHAIN*) —

The gadget chain to use for execution. This will be application specific.

- formatter (Symbol) (defaults to: *DEFAULT_FORMATTER*) —

An optional formatter to use to encapsulate the gadget chain.

Returns:

- (String)

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```

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```
# File 'lib/msf/util/dot_net_deserialization.rb', line 52

def self.generate(cmd, gadget_chain: [DEFAULT_GADGET_CHAIN](), formatter:
[DEFAULT_FORMATTER]())
  stream = self.generate_gadget_chain(cmd, gadget_chain: gadget_chain)
  self.generate_formatted(stream, formatter: formatter)
end
```

||

.generate_formatted(stream, formatter: DEFAULT_FORMATTER) ⇒ String

Take the specified serialized blob and encapsulate it with the specified formatter.

Parameters:

- stream ([Msf::Util::DotNetDeserialization::Types::SerializedStream](#)) —

The serialized stream representing the gadget chain to format into a string.

- formatter (Symbol) (*defaults to: DEFAULT_FORMATTER*) —

The formatter to use to encapsulate the serialized data blob.

Returns:

- (String)

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```

# File 'lib/msf/util/dot_net_deserialization.rb', line 66

def self.generate_formatted(stream, formatter: [DEFAULT_FORMATTER]())
  case formatter
  when :BinaryFormatter
    formatted = [Formatters]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters.html)::
    [BinaryFormatter]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/BinaryFormatte
    r.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/BinaryFormatte
    r.html#generate-class_method)(stream)
  when :JsonNetFormatter
    formatted = [Formatters]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters.html)::
    [JsonNetFormatter]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/JsonNetFormatt
    er.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/JsonNetFormatt
    er.html#generate-class_method)(stream)
  when :LosFormatter
    formatted = [Formatters]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters.html)::
    [LosFormatter]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/LosFormatter.h
    tml).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/LosFormatter.h
    tml#generate-class_method)(stream)
  when :SoapFormatter
    formatted = [Formatters]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters.html)::
    [SoapFormatter]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/SoapFormatter.
    html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/Formatters/SoapFormatter.
    html#generate-class_method)(stream)
  else
    raise NotImplementedError, 'The specified formatter is not implemented'
  end

  formatted
end

```

.generate_gadget_chain(cmd, gadget_chain: DEFAULT_GADGET_CHAIN) ⇒ [Types::SerializedStream](#)

Generate a serialized data blob using the specified gadget chain to execute the OS command. The chosen gadget chain must be compatible with the target application.

Parameters:

- cmd (String) —

The operating system command to execute. It will automatically be prefixed with “cmd /c” by the gadget chain.

- gadget_chain (Symbol) (*defaults to: DEFAULT_GADGET_CHAIN*) —

The gadget chain to use for execution.

Returns:

- ([Types::SerializedStream](#))

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```

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```
# File 'lib/msf/util/dot_net_deserialization.rb', line 110
```

```
def self.generate_gadget_chain(cmd, gadget_chain: [DEFAULT_GADGET_CHAIN]())
  case gadget_chain
  when :ClaimsPrincipal
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [ClaimsPrincipal]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/ClaimsPrinci
    pal.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/ClaimsPrinci
    pal.html#generate-class_method)(cmd)
  when :DataSet
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [DataSet]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/DataSet.html
    ).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/DataSet.html
    #generate-class_method)(cmd)
  when :DataSetTypeSpooof
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [DataSetTypeSpooof]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/DataSetTypeS
    pooof.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/DataSetTypeS
    pooof.html#generate-class_method)(cmd)
  when :ObjectDataProvider
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [ObjectDataProvider]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/ObjectDataPr
    ovider.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/ObjectDataPr
    ovider.html#generate-class_method)(cmd)
  when :TextFormattingRunProperties
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [TextFormattingRunProperties]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/TextFormatti
    ngRunProperties.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/TextFormatti
    ngRunProperties.html#generate-class_method)(cmd)
```

```

when :TypeConfuseDelegate
  stream = [GadgetChains]
  (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
  [TypeConfuseDelegate]
  (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/TypeConfuseD
  elegate.html).[generate]
  (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/TypeConfuseD
  elegate.html#generate-class_method)(cmd)
  when :WindowsIdentity
    stream = [GadgetChains]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains.html)::
    [WindowsIdentity]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/WindowsIdent
    ity.html).[generate]
    (https://docs.metasploit.com/api/Msf/Util/DotNetDeserialization/GadgetChains/WindowsIdent
    ity.html#generate-class_method)(cmd)
  else
    raise NotImplementedError, 'The specified gadget chain is not implemented'
  end

  stream
end

```

||

.get_ancestor(obj, ancestor_type, required: true) ⇒ Object

Raises:

- (RuntimeError)

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```

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```
# File 'lib/msf/util/dot_net_deserialization.rb', line 29

def self.get_ancestor(obj, ancestor_type, required: true)
  while ! (obj.nil? || obj.is_a?(ancestor_type))
    obj = obj.parent
  end

  raise RuntimeError, "Failed to find ancestor #{ancestor_type.name}" if obj.nil? &&
required

  obj
end
```

||

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